

HW 4.5.2.b: Double Angle Formulas

1. If $\sin(x) = \frac{1}{4}$ and x is in quadrant I, then find exact values for (without solving for x):

a. $\sin(2x)$

b. $\cos(2x)$

c. $\tan(2x)$

2. If $\cos(x) = \frac{3}{5}$ and x is in quadrant I, then find exact values for (without solving for x):

a. $\sin(2x)$

b. $\cos(2x)$

c. $\tan(2x)$

Evaluate each expression without a calculator.

3. $\cos^2(22.5^\circ) - \sin^2(22.5^\circ)$

4. $2\cos^2(67.5^\circ) - 1$

5. $2\sin(112.5^\circ)\cos(112.5^\circ)$

6. $\frac{2\tan(75^\circ)}{1 - \tan^2(75^\circ)}$

Simplify each expression.

7. $\cos^2(9x) - \sin^2(9x)$

8. $\cos^2(6x) - \sin^2(6x)$

9. $4\sin(8x)\cos(8x)$

10. $6\sin(5x)\cos(5x)$

Solve for all solutions on the interval $[0, 2\pi)$.

11. $6\sin(2t) + 6\sin(t) = 0$

12. $\sin(2t) + \sqrt{3}\cos(t) = 0$

13. $\cos(2\theta) = 6\cos^2(\theta) - 4$

14. $2\cos^2 x - 2\cos 2x = 1$

15. $\sin(2t) = \cos(t)$

16. $\cos(2t) = \sin(t)$

17. $\cos(6x) - \cos(3x) = 0$

18. $\sin(2x)\sin(x) = \cos(x)$

Prove the identity, Manipulate the left side to equal the right side.

19. $(\sin t - \cos t)^2 = 1 - \sin(2t)$

20. $(\sin^2 x - 1)^2 = \cos(2x) + \sin^4 x$

21. $\cos^4 x - \sin^4 x = \cos(2x)$

22. $\frac{\sin(2\theta)}{1 + \cos(2\theta)} = \tan(\theta)$

23. $\cos^2\left(\theta - \frac{\pi}{2}\right) + \cos^2(\theta) = 1$

24. $\sin\left(\theta - \frac{\pi}{2}\right)\tan(\theta) = -\sin(\theta)$

Selected Answers:

1. a. $\sin(2x) = 2 \sin x \cos x$

To find $\cos x$:

$$\sin^2 x + \cos^2 x = 1$$

$$\cos^2 x = 1 - \frac{1}{16} = \frac{15}{16}$$

$$\cos x = \pm \sqrt{\frac{15}{16}} \quad \text{Note that we need the positive root since we are told } x \text{ is in quadrant 1.}$$

$$\cos x = \frac{\sqrt{15}}{4}$$

So: $\sin(2x) = 2 \left(\frac{1}{4} \right) \left(\frac{\sqrt{15}}{4} \right) = \frac{\sqrt{15}}{8}$

b. $\cos(2x) = 2 \cos^2 x - 1$

$$= \left(2 \left(\frac{15}{16} \right) \right) - 1 = \frac{30-16}{16} = \frac{7}{8}$$

c. $\tan(2x) = \frac{\sin(2x)}{\cos(2x)} = \frac{\frac{\sqrt{15}}{8}}{\frac{7}{8}} = \frac{\sqrt{15}}{7}$

3. $\cos^2 x - \sin^2 x = \cos(2x)$, so $\cos^2(22.5^\circ) - \sin^2(22.5^\circ) = \cos(45^\circ) = \frac{\sqrt{2}}{2}$

5. $2\sin(x)\cos(x) = \sin(2x)$, so $2\sin(112.5^\circ)\cos(112.5^\circ) = \sin(225^\circ) = -\frac{\sqrt{2}}{2}$

7. $\cos^2(9x) - \sin^2(9x) = \cos(2(9x)) = \cos(18x)$

9. $4 \sin(8x) \cos(8x) = 2 (2 \sin(8x) \cos(8x)) = 2 \sin(16x)$

11. $6 \sin(2t) + 6 \sin t = 6 \cdot 2 \sin t \cos t + 6 \sin t = 6 \sin t (2 \cos t + 1)$, so we can solve

$$6 \sin t = 0 \quad \text{or} \quad \cos t = -1/2$$

$$t = 0, \pi \quad \text{or} \quad t = \frac{2\pi}{3}, \frac{4\pi}{3}$$

13. $\cos(2\theta) = 6 \cos^2 \theta - 4$, $2 \cos^2 \theta - 1 = 6 \cos^2 \theta - 4$

$$3 = 4 \cos^2 \theta, \quad \pm \sqrt{3}/2 = \cos \theta$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

15. $\sin(2t) = \cos t$

$$2 \sin t \cos t = \cos t$$

$$2 \sin t \cos t - \cos t = 0$$

$$\cos t (2 \sin t - 1) = 0$$

$$\cos t = 0 \quad \text{or} \quad 2 \sin t - 1 = 0$$

If $\cos t = 0$, then $t = \frac{\pi}{2}$ or $\frac{3\pi}{2}$. If $2 \sin t - 1 = 0$, then $\sin t = \frac{1}{2}$, so $t = \frac{\pi}{6}$ or $\frac{5\pi}{6}$. So $t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}$ or $\frac{5\pi}{6}$.

$$\begin{aligned}
 17. \quad & \cos(6x) - \cos(3x) = 0 \\
 & 2\cos^2(3x) - 1 - \cos(3x) = 0 \\
 & 2\cos^2(3x) - \cos(3x) - 1 = 0 \\
 & (2\cos(3x) + 1)(\cos(3x) - 1) = 0 \\
 & \cos(3x) = -\frac{1}{2} \text{ or } 1
 \end{aligned}$$

Since we need solutions for x in the interval $[0, 2\pi)$, we will look for all solutions for $3x$ in the interval $[0, 6\pi)$. If $\cos(3x) = -\frac{1}{2}$, then there are two possible sets of solutions. First, $3x = \frac{2\pi}{3} + 2\pi k$ where $k = 0, 1, \text{ or } 2$, so $x = \frac{2\pi}{9} + \frac{2\pi k}{3}$ where $k = 0, 1, \text{ or } 2$. Second, $3x = \frac{4\pi}{3} + 2\pi k$ where $k = 0, 1, \text{ or } 2$, so $x = \frac{4\pi}{9} + \frac{2\pi k}{3}$ where $k = 0, 1, \text{ or } 2$. If $\cos(3x) = 1$, then $3x = 2\pi k$ where $k = 0, 1, \text{ or } 2$, so $x = \frac{2\pi k}{3}$ where $k = 0, 1, \text{ or } 2$.

$$23. \cos^2\left(\theta - \frac{\pi}{2}\right) + \cos^2(\theta) = 1, \text{ substitute the cofunction identity } \sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$$

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$1 = 1$$